

REDRAG Tools — Framework Reference

Complete API Documentation for Tool Developers

Version 2.0.0 (Phase 8)

ARCHITECTURE OVERVIEW

GitHub Pages	
(Static hosting, zero build step)	
Tool Island (vram/, ram/, storage/)	
├─ index.html	→ SEO + Structure
├─ tool.css	→ <150 lines styles
├─ tool.js	→ RTTool namespace
└─ data.json	→ Tool data (optional)
Shared Shell (shared/)	
├─ css/	→ Design system
├─ js/	→ RT.* APIs
├─ html/	→ Header/Footer
└─ data/	→ tools-registry.json

RT NAMESPACE API

RT.config

```
RT.config.brand.name      // "REDRAG"
RT.config.brand.domain    // "tools.redrag.in"
RT.config.brand.blogDomain // "https://www.redrag.in"
RT.config.brand.founded   // 2026

RT.config.urls.blog       // "https://www.redrag.in/"
RT.config.urls.about      // "https://www.redrag.in/p/about-me.html"
RT.config.urls.contact    // "https://www.redrag.in/p/contact-us.html"
RT.config.urls.privacy    // "https://www.redrag.in/p/privacy-policy.html"
RT.config.urls.toolsHub   // "https://tools.redrag.in/"
RT.config.urls.sitemap    // "https://tools.redrag.in/sitemap.xml"

RT.config.paths.shared    // "/shared/"
RT.config.paths.css       // "/shared/css/"
RT.config.paths.js        // "/shared/js/"
RT.config.paths.html      // "/shared/html/"
RT.config.paths.data      // "/shared/data/"

RT.config.features.shareResults // true
```

```
RT.config.features.persistState // true
RT.config.features.analytics    // true
```

RT.dom

```
// querySelector shorthand
RT.dom.$('selector')           // Single element
RT.dom.$('selector', context)  // Within context

// querySelectorAll shorthand
RT.dom.$$('.selector')         // All matching elements
RT.dom.$$('.selector', context) // Within context

// Create element with attributes and children
RT.dom.create('div', {
  class: 'my-class',
  id: 'my-id',
  text: 'Hello',           // Sets textContent
  html: '<b>Hello</b>', // Sets innerHTML (use carefully)
  'data-value': '123',    // Any attribute
  onclick: function() {} // Event listener
}, ['child1', childElement]);

// Inject HTML string safely
RT.dom.inject('#target', '<p>HTML</p>', 'beforeend');

// Clear all children
RT.dom.clear(element);

// Check if element is in viewport
RT.dom.isInViewport(element); // Returns boolean

// DOM ready wrapper
RT.dom.ready(callback); // Runs callback when DOM is ready
```

RT.utils

```
// Debounce (wait for pause in typing)
RT.utils.debounce(function, delayMs);
// Usage: RT.utils.debounce(() => calculate(), 150)

// Throttle (limit execution rate)
RT.utils.throttle(function, limitMs);

// XSS-safe HTML escaping
RT.utils.escapeHtml('<script>alert(1)</script>');
// Returns: &lt;script&gt;alert(1)&lt;/script&gt;
```

```

// XSS-safe JS string escaping
RT.utils.escapeJs("It's a test");

// Copy to clipboard
RT.utils.copyToClipboard('text').then(success => {
  if (success) console.log('Copied!');
});

// Format number with Indian locale
RT.utils.formatNumber(1000000); // "10,00,000"

// Format bytes to human-readable
RT.utils.formatBytes(1073741824); // "1.00 GB"

// URL query parameters
RT.utils.getQueryParam('key'); // Get single param
RT.utils.setQueryParams({key: 'value'}); // Set multiple params (no reload)

// localStorage wrapper (auto-prefixed with "rt:")
RT.utils.storage.get('key', defaultValue);
RT.utils.storage.set('key', value);
RT.utils.storage.remove('key');

// Device detection
RT.utils.prefersReducedMotion(); // Boolean
RT.utils.isTouchDevice(); // Boolean

```

RT.components

```

// Accordion (FAQ-style expand/collapse)
RT.components.Accordion.init(container, {
  multiple: false, // Allow multiple open?
  animated: true // Animate transitions?
});

// Tabs
RT.components.Tabs.init(container);

// Toast notification
RT.components.Toast.show({
  message: 'Hello!',
  type: 'success', // info | success | error | warning
  duration: 3000, // ms (0 = persistent)
  dismissible: true,
  action: {
    label: 'Undo',
    onClick: () => {}
  }
});

```

```

// Tooltip
RT.components.Tooltip.attach(element, {
  content: 'Tooltip text',
  position: 'top',    // top | bottom | left | right
  delay: 300
});

// Modal
const modal = RT.components.Modal.open({
  title: 'Title',
  content: '<p>HTML content</p>',
  size: 'md',        // sm | md | lg | full
  closable: true,
  showFooter: true,
  footerButtons: [
    { label: 'OK', variant: 'primary', onClick: (e, modal) => modal.close() },
    { label: 'Cancel', variant: 'secondary', onClick: (e, modal) => modal.close() }
  ]
});

// Spinner
RT.components.Spinner.create({ size: 'medium', color: 'primary' });

// Skeleton placeholder
RT.components.Skeleton.create({ variant: 'text', width: '100%', height: '16px' });

// Search input
const search = RT.components.Search.create({
  placeholder: 'Search...',
  debounce: 200,
  onSearch: (query) => console.log(query)
});
search.search('term'); // Programmatic search
search.clear();        // Clear input

// FAQ with auto schema injection
RT.components.FAQ.init({
  container: '#faq-container',
  items: [
    { question: 'Q1?', answer: '<p>A1</p>' },
    { question: 'Q2?', answer: '<p>A2</p>' }
  ],
  showSchema: true,    // Auto-inject FAQPage schema
  openFirst: false    // Open first item by default?
});

// Create button element
RT.components.createButton({
  text: 'Click me',
  variant: 'primary',  // primary | secondary | ghost | danger

```

```

    size: 'md',          // sm | md | lg
    icon: '→',
    iconPosition: 'right',
    fullWidth: false,
    disabled: false,
    loading: false,
    ariaLabel: 'Click button',
    onClick: () => {}
  });

  // Create card element
  RT.components.createCard({
    variant: 'default',
    padding: '24px',
    border: true,
    hover: true,
    accent: 'top',       // top | left
    href: '/link/',
    ariaLabel: 'Card',
    children: [element, 'text']
  });

```

RT.seo

```

// Inject any schema
RT.seo.injectSchema({
  '@context': 'https://schema.org',
  '@type': 'YourSchemaType',
  // ... properties
}, 'unique-id');

// Pre-built schema generators
RT.seo.breadcrumbSchema([
  { name: 'Home', item: 'https://www.redrag.in/' },
  { name: 'Tools', item: 'https://tools.redrag.in/' },
  { name: 'Your Tool', item: 'https://tools.redrag.in/your-tool/' }
]);

RT.seo.softwareApplicationSchema({
  name: 'Tool Name',
  description: 'Description',
  url: 'https://tools.redrag.in/your-tool/',
  ratingValue: '4.8',
  ratingCount: '100'
});

RT.seo.faqSchema([
  { question: 'Q?', answer: 'A' }
]);

```

```

RT.seo.howToSchema({
  name: 'How to...',
  description: '...',
  totalTime: 'PT2M',
  steps: [
    { name: 'Step 1', text: '...', url: '...#step1' }
  ],
  supplies: ['Item 1'],
  tools: ['Tool 1']
});

RT.seo.articleSchema({
  headline: 'Article Title',
  description: '...',
  image: '...',
  datePublished: '2026-07-31',
  dateModified: '2026-07-31',
  url: '...',
  wordCount: 1500,
  articleSection: 'Gaming Tools',
  mainEntity: { '@id': '...#webapp' }
});

RT.seo.websiteSchema(); // Auto-uses RT.config.urls
RT.seo.organizationSchema(); // Auto-uses RT.config.urls

RT.seo.itemListSchema([
  { name: 'Tool 1', slug: 'tool-1', description: '...' },
  { name: 'Tool 2', slug: 'tool-2', description: '...' }
]);

// Meta tag helpers
RT.seo.setMeta('name', 'content'); // name attribute
RT.seo.setMeta('og:title', 'content', 'property'); // property attribute

RT.seo.setOpenGraph({
  title: '...',
  description: '...',
  url: '...',
  image: '...',
  imageWidth: 1200,
  imageHeight: 630,
  imageAlt: '...',
  siteName: 'REDRAG Tools',
  type: 'website',
  locale: 'en_US'
});

RT.seo.setTwitterCard({
  title: '...',

```

```

description: '...',
image: '...',
imageAlt: '...',
card: 'summary_large_image',
site: '@redragblog',
creator: '@redragblog',
label1: 'Label',
data1: 'Value',
label2: 'Label',
data2: 'Value'
});

// Utility
RT.seo.stripHtml('<p>Text</p>'); // Returns: "Text"
RT.seo.getCanonicalUrl(); // Returns current canonical URL

```

RT.shell

```

// Inject header (auto-called by RT.init())
RT.shell.injectHeader({ activeNav: 'tools' });

// Inject footer (auto-called by RT.init())
RT.shell.injectFooter();

// Wait for both to be ready
RT.shell.ready.then(() => {
  console.log('Header and footer loaded');
});

```

RT.analytics

```

// Track custom event
RT.analytics.track('Event Name', { prop: 'value' });

// Track tool action (recommended)
RT.analytics.trackTool('tool-slug', 'action');
// Actions: load, calculate, share, reset, cta_click

```

DESIGN TOKENS (CSS Variables)

Use these in your tool.css:

```

/* Brand Colors */
var(--rt-primary)      /* #e63946 */
var(--rt-primary-light) /* #ff6b7a */
var(--rt-primary-dark) /* #c1121f */

```

```
/* Backgrounds */
var(--rt-bg-base)          /* #0a0a0f */
var(--rt-bg-elevated)      /* #111118 */
var(--rt-bg-card)          /* #16161f */
var(--rt-bg-card-hover)    /* #1e1e2a */
var(--rt-bg-input)         /* #0f0f16 */
var(--rt-bg-overlay)       /* rgba(10,10,15,0.85) */

/* Borders */
var(--rt-border-default)   /* #252535 */
var(--rt-border-hover)     /* #353550 */
var(--rt-border-focus)     /* #e63946 */

/* Text */
var(--rt-text-primary)     /* #f0f0f5 */
var(--rt-text-secondary)   /* #a0a0b0 */
var(--rt-text-muted)       /* #7a7a8a */
var(--rt-text-disabled)    /* #4a4a5a */

/* Semantic */
var(--rt-success)          /* #22c55e */
var(--rt-warning)          /* #f59e0b */
var(--rt-danger)           /* #ef4444 */
var(--rt-info)             /* #3b82f6 */

/* Typography */
var(--rt-text-hero)        /* clamp(36px, 6vw, 64px) */
var(--rt-text-h1)          /* clamp(28px, 4vw, 40px) */
var(--rt-text-h2)          /* clamp(24px, 3.5vw, 32px) */
var(--rt-text-h3)          /* 20px */
var(--rt-text-h4)          /* 17px */
var(--rt-text-body)        /* 15px */
var(--rt-text-body-lg)     /* 17px */
var(--rt-text-small)       /* 13px */
var(--rt-text-xs)          /* 11px */

/* Font Weights */
var(--rt-weight-regular)   /* 400 */
var(--rt-weight-medium)    /* 500 */
var(--rt-weight-semibold)  /* 600 */
var(--rt-weight-bold)      /* 700 */
var(--rt-weight-extrabold) /* 800 */
var(--rt-weight-black)    /* 900 */

/* Spacing (8px base) */
var(--rt-space-1)          /* 4px */
var(--rt-space-2)          /* 8px */
var(--rt-space-3)          /* 12px */
var(--rt-space-4)          /* 16px */
var(--rt-space-5)          /* 24px */
```



```

var(--rt-space-6) /* 32px */
var(--rt-space-8) /* 48px */
var(--rt-space-10) /* 64px */
var(--rt-space-12) /* 80px */

/* Border Radius */
var(--rt-radius-sm) /* 8px */
var(--rt-radius) /* 12px */
var(--rt-radius-lg) /* 16px */
var(--rt-radius-xl) /* 20px */
var(--rt-radius-full)/* 9999px */

/* Shadows */
var(--rt-shadow-sm) /* 0 2px 8px rgba(0,0,0,0.3) */
var(--rt-shadow) /* 0 4px 24px rgba(0,0,0,0.4) */
var(--rt-shadow-lg) /* 0 12px 48px rgba(0,0,0,0.5) */
var(--rt-shadow-glow) /* 0 4px 16px rgba(230,57,70,0.3) */
var(--rt-shadow-glow-lg)/* 0 8px 32px rgba(230,57,70,0.4) */

/* Z-Index */
var(--rt-z-dropdown) /* 50 */
var(--rt-z-sticky) /* 100 */
var(--rt-z-overlay) /* 200 */
var(--rt-z-modal) /* 300 */
var(--rt-z-toast) /* 400 */

/* Layout */
var(--rt-container-max) /* 1280px */
var(--rt-header-height) /* 72px */

```

CSS COMPONENT CLASSES

Buttons

```

<button class="rt-btn rt-btn--primary">Primary</button>
<button class="rt-btn rt-btn--secondary">Secondary</button>
<button class="rt-btn rt-btn--ghost">Ghost</button>
<button class="rt-btn rt-btn--danger">Danger</button>
<button class="rt-btn rt-btn--filter is-active">Filter</button>
<button class="rt-btn rt-btn--icon"></button>
<button class="rt-btn rt-btn--sm">Small</button>
<button class="rt-btn rt-btn--lg">Large</button>
<button class="rt-btn rt-btn--full">Full Width</button>

```

Cards

```

<div class="rt-card">Default</div>
<div class="rt-card rt-card--hover">Hover Lift</div>
<div class="rt-card rt-card--accent-top">Top Accent</div>
<div class="rt-card rt-card--accent-left is-selected">Left Accent</div>

```

```
<div class="rt-card rt-card--feature">Feature</div>
<div class="rt-card rt-card--stat">Stat</div>
<div class="rt-card rt-card--result">Result</div>
<div class="rt-card rt-card--skeleton">Skeleton</div>
```

Inputs

```
<input class="rt-input" placeholder="Text">
<input class="rt-input rt-input--search" placeholder="Search">
<input class="rt-input rt-input--number" type="number">
```

Tags

```
<span class="rt-tag rt-tag--live">● LIVE</span>
<span class="rt-tag rt-tag--coming">Coming Soon</span>
<span class="rt-tag rt-tag--neutral">Neutral</span>
<span class="rt-tag rt-tag--primary">Primary</span>
```

Progress Bar

```
<div class="rt-progress">
  <div class="rt-progress__label"><span>Label</span><span>50%</span></div>
  <div class="rt-progress__track">
    <div class="rt-progress__fill rt-progress__fill--safe" style="width:50%">50%</div>
  </div>
</div>
```

Insight Box

```
<div class="rt-insight">
  <div class="rt-insight__title">Title</div>
  <div>Content</div>
</div>
<div class="rt-insight rt-insight--danger">
  <div class="rt-insight__title">Warning</div>
  <div>Content</div>
</div>
```

Tables

```
<div class="rt-table-wrap">
  <table class="rt-table">
    <thead><tr><th>Col 1</th><th>Col 2</th></tr></thead>
    <tbody><tr><td>Data</td><td>Data</td></tr></tbody>
  </table>
</div>
```

ACCESSIBILITY REQUIREMENTS

Every tool MUST have:

```
<!-- Skip link -->
<a href="#rt-main" class="rt-skip-link">Skip to main content</a>

<!-- Main landmark -->
<main id="rt-main" role="main" aria-label="Tool Name Calculator">

<!-- Section labels -->
<section aria-label="Tool introduction">
<div role="complementary" aria-label="Calculation results">
<div role="form" aria-label="Calculator inputs">
<div role="region" aria-label="Calculation details">

<!-- Live regions for dynamic content -->
<div aria-live="polite" aria-atomic="false">

<!-- Progress bar -->
<div role="progressbar" aria-valuemin="0" aria-valuemax="100" aria-valuenow="0">

<!-- Buttons -->
<button aria-label="Descriptive action">Label</button>

<!-- Inputs -->
<input aria-label="Input description">
<select aria-label="Select description">
```

Keyboard support:

- Tab navigation through all interactive elements
- Enter/Space to activate buttons
- Escape to close modals/menus
- Ctrl+R to reset (template mein built-in)
- Ctrl+C to copy results (template mein built-in)

SECURITY RULES

NEVER DO:

```
// ❌ Inline onclick (XSS vulnerability)
<button onclick="calculate()">Click</button>

// ❌ innerHTML with user input
el.innerHTML = userInput;

// ❌ Hardcoded URLs
<a href="https://tools.redrag.in/vram/">
```

```
// ❌ alert()/confirm()
alert('Error');
```

ALWAYS DO:

```
// ✅ Event delegation
container.addEventListener('click', handleClick);

// ✅ Escape HTML
el.textContent = RT.utils.escapeHtml(userInput);

// ✅ Use RT.config.urls
<a href="${RT.config.urls.toolsHub}vram/">

// ✅ Use RT.components.Toast
RT.components.Toast.show('Error', 'error', 3000);
```

📐 PERFORMANCE CHECKLIST

- Tool CSS < 150 lines
- No duplicated shared styles
- ?v=2 on shared assets, ?v=1 on tool assets
- content-visibility: auto on below-fold sections (inherited from core.css)
- Images have loading="lazy" (if any)
- No inline styles (use CSS classes)
- No inline onclick (event delegation)
- Data loaded via fetch() (async, non-blocking)

🔧 TESTING COMMANDS

Browser Console (F12):

```
// Check if RT loaded
console.log(RT.version); // Should print "2.0.0"

// Check if tool loaded
console.log(RTTool.meta); // Should print tool info

// Check schemas
console.log(document.querySelectorAll('script[type="application/ld+json"]'));

// Test localStorage
RT.utils.storage.set('test', {a: 1});
console.log(RT.utils.storage.get('test'));

// Test clipboard
```

```
RT.utils.copyToClipboard('test').then(s => console.log(s));
```

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For questions, see `TOOL_CREATION_GUIDE.md`